

● HARD

● ALGEBRA

● ANSWER KEY

Algebra

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03

Q1**1**

The correct answer is 1.25. The y-coordinate of the x-intercept is 0, so 0 can be substituted for y, giving $\frac{4}{5}x + \frac{1}{3}(0) = 1$. This simplifies to $\frac{4}{5}x = 1$. Multiplying both sides of $\frac{4}{5}x = 1$ by 5 gives $4x = 5$. Dividing both sides of $4x = 5$ by 4 gives $x = \frac{5}{4}$, which is equivalent to 1.25. Note that 1.25 and 5/4 are examples of ways to enter a correct answer.

Q2**C**

C. $\frac{61c}{4} + 105$

Choice C is correct. It's given that $gc + 7 = \frac{c}{4}$. Therefore, for the given linear function g, when $x = c + 7$, $gx = \frac{c}{4}$. Substituting $c + 7$ for x and $\frac{c}{4}$ for gx in $gx = b - 15x$ yields $\frac{c}{4} = b - 15c + 7$. Applying the distributive property to the right-hand side of this equation yields $\frac{c}{4} = b - 15c - 105$. Adding $15c$ to both sides of this equation yields $\frac{c}{4} + 15c = b - 105$. Adding 105 to both sides of this equation yields $\frac{c}{4} + 15c + 105 = b$, or $\frac{61c}{4} + 105 = b$. Therefore, the expression that represents the value of b is $\frac{61c}{4} + 105$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**C**

C. 22

Choice C is correct. Ken earned \$8 per hour for the first 10 hours he worked, so he earned a total of \$80 for the first 10 hours he worked. For the rest of the week, Ken was paid at the rate of \$10 per hour. Let x be the number of hours he will work for the rest of the week. The total of Ken's earnings, in dollars, for the week will be $10x + 80$. He saves 90% of his earnings each week, so this week he will save $0.9(10x + 80)$ dollars. The inequality $0.9(10x + 80) \geq 270$ represents the condition that he will save at least \$270 for the week. Factoring 10 out of the expression $10x + 80$ gives $10(x + 8)$. The product of 10 and 0.9 is 9, so the inequality can be rewritten as $9(x + 8) \geq 270$. Dividing both sides of this inequality by 9 yields $x + 8 \geq 30$, so $x \geq 22$. Therefore, the least number of hours Ken must work the rest of the week to save at least \$270 for the week is 22.

Choices A and B are incorrect because Ken can save \$270 by working fewer hours than 38 or 33 for the rest of the week. Choice D is incorrect. If Ken worked 16 hours for the rest of the week, his total earnings for the week will be $\$80 + \$160 = \$240$, which is less than \$270. Since he saves only 90% of his earnings each week, he would save even less than \$240 for the week.

Q4**.9411**

Accept also: .9412, 16/17

The correct answer is $\frac{16}{17}$. It's given that the equation $3kx + 13 = \frac{48}{17}x + 36$ has no solution. A linear equation in the form $ax + b = cx + d$, where a , b , c , and d are constants, has no solution only when the coefficients of x on each side of the equation are equal and the constant terms aren't equal. Dividing both sides of the given equation by 3 yields $kx + 13 = \frac{48}{51}x + \frac{36}{3}$, or $kx + 13 = \frac{16}{17}x + 12$. Since the coefficients of x on each side of the equation must be equal, it follows that the value of k is $\frac{16}{17}$. Note that 16/17, .9411, .9412, and 0.941 are examples of ways to enter a correct answer.

Q5**.2857**

Accept also: 2/7

The correct answer is $\frac{2}{7}$. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions if and only if the two linear equations are equivalent. Multiplying each side of the first equation in the system by $\frac{35}{4}$ yields $\frac{35}{4}x + \frac{7}{5}y = \frac{35}{4}$, or $\frac{7}{2}x + \frac{49}{4}y = \frac{5}{2}$. Since this equation is equivalent to the second equation and has the same right side as the second equation, the coefficients of x and y , respectively, should also be the same. It follows that $g = \frac{7}{2}$ and $k = \frac{49}{4}$. Therefore, the value of $\frac{g}{k}$ is $\frac{\frac{7}{2}}{\frac{49}{4}}$, or $\frac{2}{7}$. Note that 2/7, .2857, 0.285, and 0.286 are examples of ways to enter a correct answer.

Q6**A****A. 811**

Choice A is correct. The x -intercept of the graph shown represents the situation in which the cluster contains only M-type stars. Based on the graph, the x -intercept is approximately (158, 0). Therefore, if there are 0 K-type stars in the cluster, then there are approximately 158 M-type stars in the cluster. It's given that the total mass of the stars in this cluster is 127,882 quettagrams. Therefore, the estimated mass of each M-type star is approximately $\frac{127,882}{158}$ quettagrams, or approximately 809.38 quettagrams. Thus, of the given choices, 811 is the closest to the estimated mass, in quettagrams, of each M-type star in this cluster.

Choice B is incorrect. This is the value closest to the estimated mass, in quettagrams, of each K-type star, not each M-type star, in this cluster.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**A**

A. When $f(x) = 0$, the number is $-\frac{19}{4}$.

Choice A is correct. It's given that the function $f(x)$ is defined as 19 more than 4 times a number x . This can be represented by the equation $f(x) = 4x + 19$. The x -intercept of the graph of $y = f(x)$ in the xy -plane is the point where the graph intersects the x -axis, or the point on the graph where the value of $f(x)$ is equal to 0. Substituting 0 for $f(x)$ in the equation $f(x) = 4x + 19$ yields $0 = 4x + 19$. Subtracting 19 from each side of this equation yields $-19 = 4x$. Dividing each side of this equation by 4 yields $x = -\frac{19}{4}$. Therefore, when $f(x) = 0$, the number is $-\frac{19}{4}$.

Choice B is incorrect. This is the best interpretation of the y -intercept, not the x -intercept, of the graph of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the best interpretation of the slope, not the x -intercept, of the graph of the function.

Q8**B****B.** $y > 4$

Choice B is correct. Subtracting the same number from each side of an inequality gives an equivalent inequality. Hence, subtracting 1 from each side of the inequality $2x > 5$ gives $2x - 1 > 4$. So the given system of inequalities is equivalent to the system of inequalities $y > 2x - 1$ and $2x - 1 > 4$, which can be rewritten as $y > 2x - 1 > 4$. Using the transitive property of inequalities, it follows that $y > 4$.

Choice A is incorrect because there are points with a y-coordinate less than 6 that satisfy the given system of inequalities. For example, $(3, 5.5)$ satisfies both inequalities. Choice C is incorrect. This may result from solving the inequality $2x > 5$ for x, then replacing x with y. Choice D is incorrect because this inequality allows y-values that are not the y-coordinate of any point that satisfies both inequalities. For example, $y = 2$ is contained in the set $y > \frac{3}{2}$;

however, if 2 is substituted into the first inequality for y, the result is $x < \frac{3}{2}$. This cannot be true because the second inequality gives $x > \frac{5}{2}$.

Q9**D**

D. $6n + 22 = 100$

Choice D is correct. It's given that during a certain day at a factory, the number of 7-inch concrete screws the factory makes is n and the number of 4-inch concrete screws the factory makes is 22. It's also given that during this day the number of 9-inch concrete screws the factory makes is 5 times the number of 7-inch concrete screws, or $5n$. Therefore, the total number of 7-inch, 9-inch, and 4-inch concrete screws is $n + 5n + 22$, or $6n + 22$. It's given that during this day, the factory makes 100 concrete screws total. Thus, the equation $6n + 22 = 100$ represents this situation.

Choice A is incorrect. This equation represents a situation where the total length, in inches, of all the concrete screws, rather than the total number of concrete screws, is 100.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the total number of 9-inch concrete screws and 4-inch concrete screws, not including the 7-inch concrete screws, is 100.

Q10

B

B. $-\frac{3r}{2} + \frac{7}{2}, r$

Choice B is correct. The two given equations are equivalent because the second equation can be obtained from the first equation by multiplying each side of the equation by 5. Thus, the graphs of the equations are coincident, so if a point lies on the graph of one of the equations, it also lies on the graph of the other equation. A point x, y lies on the graph of an equation in the xy -plane if and only if this point represents a solution to the equation. It is sufficient, therefore, to find the point that represents a solution to the first given equation. Substituting the x - and y -coordinates of choice B, $-\frac{3r}{2} + \frac{7}{2}$ and r , for x and y , respectively, in the first equation yields $2(-\frac{3r}{2} + \frac{7}{2}) + 3r = 7$, which is equivalent to $-3r + 7 + 3r = 7$, or $7 = 7$. Therefore, the point $-\frac{3r}{2} + \frac{7}{2}, r$ represents a solution to the first equation and thus lies on the graph of each equation in the xy -plane for the given system.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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